Chapter 3: Relational Algebra

Content

1. Introduction
2. Relational algebra expressions
3. Relational algebra operations
4. Example exercises

1. Introduction

- Relational Algebra provides the mathematical foundation (specifically, set theory) for modeling relational databases. The objects of manipulation are the relations in a relational database.
- **Functions:**
  - Allow the description of operations for extracting data from relations in a relational database.
  - Allow optimization of the extraction process using the available operations of set theory.

2. Relational algebra expressions

- A relational algebra expression is an expression composed of relational algebra operations.
- A relational algebra expression is considered as a relation (without a name).
- It is possible to assign a name to the relation created from a relational algebra expression.
- It is possible to rename the attributes of the relation created from a relational algebra expression.

3. Relational algebra operations

3.1 Introduction
3.2 Selection
3.3 Projection
3.4 Assignment
3.5 Set Operations
3.6 Join
3.7 Division
3.8 Aggregate Functions and Grouping

3.1 Introduction (1)

- There are five basic operations:
  - **Selection (σ) or (:)**
  - **Projection (π) or ([])**
  - **Cartesian Product (×)**
  - **Difference (–)**
  - **Union (∪)**

3.1 Introduction (2)

- Other operations that are not basic but useful:
  - Intersection ($\cap$)
  - Join ($\bowtie$)
  - Division ($\div$)
  - Complement ($\neg$)
  - Renaming (ϱ)
  - Assignment ($\leftarrow$)
- The result of performing these operations is always a relation, which means they can be combined to form new operations.

3.2 Selection

- Extract tuples (rows) from relation R. The extracted tuples must satisfy **the selection condition p**.
- **Symbol:** $\sigma_p(R)$
- **Definition:** $\sigma_p(R)=\{t/t\in R, p(t)\}$
- The returned result is a relation, with the same attribute list as relation R. No duplicate rows in the results.
- Selection is commutative:

$$\sigma_{p1}(\sigma_{p2}(R))=\sigma_{p2}(\sigma_{p1}(R))=\sigma_{(p1\wedge p2)}(R)$$

Academic management database diagram

HOCVIEN (MAHV, HO, TEN, NGSINH, GIOITINH, NOISINH, MALOP)

LOP (MALOP, TENLOP, TRGLOP, SISO, MAGVCN)

KHOA (MAKHOA, TENKHOA, NGTLAP, TRGKHOA)

MONHOC (MAMH, TENMH, TCLT, TCTH, MAKHOA)

DIEUKIEN (MAMH, MAMH_TRUOC)

GIAOVIEN(MAGV,HOTEN,HOCVI,HOCHAM,GIOITINH,NGSINH,NGVL, HESO, MUCLUONG, MAKHOA)

GIANGDAY(MALOP,MAMH,MAGV,HOCKY, NAM,TUNGAY,DENNGAY)

KETQUATHI (MAHV, MAMH, LANTHI, NGTHI, DIEM, KQUA)

3.2 Selection example

- Looking for male students born in Ho Chi Minh City

$$\sigma_{(Gioitinh='Nam') \wedge (Noisinh='TpHCM')}(HOCVIEN)$$

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11
K1105	Tran Minh Long	Nam	TpHCM	K11
K1106	Le Nhat Minh	Nam	TpHCM	K11

3.3 Projection

- Used to extract the values of some attributes (columns) of a relation *R*.
- **Symbol:** $\pi_{A_1, A_2, \ldots, A_k}(R)$

where A_i are the names of the projected attributes.

- The result returns a relation with k attributes in the order listed. No duplicate rows in the results.
- Projection is not commutative.

3.3 Examples

- Find the ID and full name of male students born in Ho Chi Minh City

$$\pi_{Mahv,Hoten}\sigma_{(Gioitinh='Nam') \wedge (Noisinh='TpHCM')}(HOCVIEN)$$

HOCVIEN				
Mahv	**HoTen**	**Gioitinh**	**Noisinh**	**Malop**
K1103	Ha Duy Lap	Nam	Nghe An	K11
K1102	Tran Ngoc Han	Nu	Kien Giang	K11
K1104	Tran Ngoc Linh	Nu	Tay Ninh	K11
K1105	Tran Minh Long	Nam	TpHCM	K11
K1106	Le Nhat Minh	Nam	TpHCM	K11

3.4 Assignment

- Used when expressing complex queries.
- **Symbol:** $A \leftarrow B$
- **For example:**
  $R(NAME, SALARY) \leftarrow \pi_{NAME, SALARY}(EMPLOYEE)$
- The result of the right-hand side is assigned to the relation variable on the left-hand side.

3.5 Set operations

3.5.1 Introduction

3.5.2 Union ($\cup$)

3.5.3 Difference (–)

3.5.4 Intersection ($\cap$)

3.5.5 Cartesian Product ($\times$)

3.5.1 Introduction

- The operations performed on two relations come from set theory in mathematics: union ($R \cup S$), intersection ($R \cap S$), difference ($R - S$), and Cartesian product ($R \times S$).
- For union, intersection, and difference, the relations R and S must be ***compatible***:
  - The number of attributes of R and S must be the same: $R(A_1, A_2, \ldots A_n)$ and $S(B_1, B_2, \ldots B_n)$
  - The value domains of the attributes must be compatible: $dom(A_i)=dom(B_i)$
- For union, intersection, and difference, the resulting relation has the same attribute names as the first relation.

3.5.2 Union

- **Symbol:** R ∪ S
- **Definition:** $R \cup S = \{t \mid t \in R \vee t \in S\}$

  where R and S are two compatible relations.
- **Example:** Students awarded in round 1 or round 2.

DOT1	
Mahv	**Hoten**
K1103	Le Van Tam
K1114	Tran Ngoc Han
K1203	Le Thanh Hau
K1308	Nguyen Gia

DOT2	
Mahv	**Hoten**
K1101	Le Kieu My
K1114	Tran Ngoc Han

Mahv	Hoten
K1101	Le Kieu My
K1103	Le Van Tam
K1114	Tran Ngoc Han
K1203	Le Thanh Hau
K1308	Nguyen Gia

DOT1 ∪ DOT2

3.5.3 Difference

- **Symbol:** RS
- **Definition:** $R - S = \{t \mid t \in R \wedge t \notin S\}$

  where R and S are two compatible relations.
- **For example:** Students awarded in round 1 but not in round 2.

DOT1	
Mahv	**Hoten**
K1103	Le Van Tam
K1114	Tran Ngoc Han
K1203	Le Thanh Hau
K1308	Nguyen Gia

DOT2	
Mahv	**Hoten**
K1101	Le Kieu My
K1114	Tran Ngoc Han

Mahv	Hoten
K1103	Le Van Tam
K1203	Le Thanh Hau
K1308	Nguyen Gia

DOT1 – DOT2

3.5.4 Intersection

- **Symbol:** R∩S
- **Definition:** $R \cap S = \{t \mid t \in R \wedge t \in S\}$

  where R, S are two compatible relations.
- Other definition: $R \cap S = R - (R - S)$
- **Example:** Students awarded in both round 1 and round 2.

KT_D1	
Mahv	**Hoten**
K1103	Le Van Tam
K1114	Tran Ngoc Han
K1203	Le Thanh Hau
K1308	Nguyen Gia

KT_D2	
Mahv	**Hoten**
K1101	Le Kieu My
K1114	Tran Ngoc Han

Mahv	Hoten
K1114	Tran Ngoc Han

DOT1 ∩ DOT2

3.5.5 Cartesian Product (1)

- **Symbol:** R×S
- **Definition:** $R \times S = \{t_r t_s / t_r \in R \wedge t_s \in S\}$
- If R has n tuples and S has m tuples, the result will have n*m tuples, and the number of attributes equal to the total number of attributes from R and S.
- The Cartesian product is often used in combination with selection to join related tuples from two relations.
- **Example:** From 2 relations HOCVIEN and MONHOC, find all possible cases of *"a student registered for a course,"* assuming there are no conditions at all.

3.5.5 Cartesian Product (2)

HOCVIEN	
Mahv	**Hoten**
K1103	Le Van Tam
K1114	Tran Ngoc Han
K1203	Le Thanh Hau

MONHOC
Mamh
CTRR
THDC
CTDL

Mahv	Hoten	Mamh
K1103	Le Van Tam	CTRR
K1114	Tran Ngoc Han	CTRR
K1203	Le Thanh Hau	CTRR
K1103	Le Van Tam	THDC
K1114	Tran Ngoc Han	THDC
K1203	Le Thanh Hau	THDC
K1103	Le Van Tam	CTDL
K1114	Tran Ngoc Han	CTDL
K1203	Le Thanh Hau	CTDL

HOCVIEN×MONHOC

3.6 Join

3.6.1 Join

3.6.2 Equi-join, Natural join

3.6.3 Outer join

3.6.1 Theta-Join (1)

- Theta-join (θ): Similar to Cartesian product combined with selection. The selection condition is called the join condition.
- **Symbol:** $R \overset{p}{\bowtie} S$

  where R,S are relations, p is the condition of the connection
- Tuples with NULL values in the join attributes do not appear in the join result.
- A join with a general condition is called a θ-join, with θ being one of the comparison operators (≠, =, >, ≥, <, ≤).

3.6.1 Theta-Join (2)

$$R \bowtie_{A_1 > B_2} S$$

R	
A_1	A_2
1	2
1	8
0	0
8	4
0	3

S		
B_1	B_2	B_3
0	2	8
7	8	7
8	0	4
1	0	7
2	1	5

A_1	A_2	B_1	B_2	B_3
1	2	8	0	4
1	2	1	0	7
1	8	8	0	4
1	8	1	0	7
8	4	0	2	8
8	4	8	0	4
8	4	1	0	7
8	4	2	1	5

3.6.2 Equi-join and Natural join

- If θ is the equality operator (=), the join is called an equi-join.

  <u>Symbol:</u> $HOCVIEN \overset{Mahv=Trglop}{\bowtie} LOP$

- If the equi-join condition is the same attributes, it is called a natural-join. In this case, the result eliminates one of the two duplicate columns.

  <u>Symbol:</u> $HOCVIEN \overset{Mahv}{\bowtie} KETQUATHI$

  or $HOCVIEN * KETQUATHI$

3.6.3 Outer join

- Expand the join to avoid information loss
- Perform the join, then add to the result those tuples from one relation that do not match with any tuple in the other relation.
- There are 3 types:
  - Left outer join R ⟕ S
  - Right outer join R ⟖ S
  - Full outer join R ⟗ S
- **Example:** Print a list of all students and their exam scores (if any exists).

Left outer join

- HOCVIEN $\overset{mahv}{\bowtie}$ KETQUATHI

Mahv	Hoten	Mahv	Mamh	Diem
HV01	Nguyen Van Lan	HV01	CSDL	7.0
HV01	Nguyen Van Lan	HV01	CTRR	8.5
HV02	Tran Hong Son	HV02	CSDL	8.5
HV03	Nguyen Le	HV03	CTRR	9.0
HV04	Le Minh	Null	Null	Null

HOCVIEN	
Mahv	**Hoten**
HV01	Nguyen Van Lan
HV02	Tran Hong Son
HV03	Nguyen Le
HV04	Le Minh

KETQUATHI		
Mahv	**Mamh**	**Diem**
HV01	CSDL	7.0
HV02	CSDL	8.5
HV01	CTRR	8.5
HV03	CTRR	9.0

Right outer join

- HOCVIEN $\overset{\text{Mahv}}{\bowtie}$ KETQUATHI

Mahv	Hoten	Mahv	Mamh	Diem
HV01	Nguyen Van Lan	HV01	CSDL	7.0
HV01	Nguyen Van Lan	HV01	CTRR	8.5
HV02	Tran Hong Son	HV02	CSDL	8.5
HV03	Nguyen Le	HV03	CTRR	9.0
Null	Null	HV05	CTRR	8.0

HOCVIEN	
Mahv	**Hoten**
HV01	Nguyen Van Lan
HV02	Tran Hong Son
HV03	Nguyen Le
HV04	Le Minh

KETQUATHI		
Mahv	**Mamh**	**Diem**
HV01	CSDL	7.0
HV02	CSDL	8.5
HV01	CTRR	8.5
HV03	CTRR	9.0
HV05	CTRR	8.0

Full outer join

- HOCVIEN $\overset{Mahv}{\bowtie}$ KETQUATHI

Mahv	Hoten	Mahv	Mamh	Diem
HV01	Nguyen Van Lan	HV01	CSDL	7.0
HV01	Nguyen Van Lan	HV01	CTRR	8.5
HV02	Tran Hong Son	HV02	CSDL	8.5
HV03	Nguyen Le	HV03	CTRR	9.0
HV04	Le Minh	Null	Null	Null
Null	Null	HV05	CTRR	8.0

HOCVIEN	
Mahv	**Hoten**
HV01	Nguyen Van Lan
HV02	Tran Hong Son
HV03	Nguyen Le
HV04	Le Minh

KETQUATHI		
Mahv	**Mamh**	**Diem**
HV01	CSDL	7.0
HV02	CSDL	8.5
HV01	CTRR	8.5
HV03	CTRR	9.0
HV05	CTRR	8.0

3.7 Division

- Find the values in R that are associated with all values in S.
- **Definition:** $Q = R \div S = \{t / \forall s \in S, \ (t, s) \in R\}$
  - For example:
    - Which student took all the subjects?
    - Which customers purchased all products?
- R and S are two relations, with R^+ and S^+ being the attribute sets of R and S, respectively. The condition is $S^+ \neq \varnothing$ and $S^+ \subseteq R^+$. Q is the result of dividing R by S, with $Q^+ = R^+ - S^+$.

3.7 Division (cont.)

- Example

R	A	B	C	D	E
	α	a	1	a	1
	α	a	2	a	1
	2	2	2	b	1
	β	b	3	a	1
	β	b	3	b	3
	γ	c	4	a	1
	γ	c	4	b	1
	γ	a	5	b	1

S	D	E
	a	1
	b	1

R ÷ S

A	B	C
α	a	2
γ	c	4

3.7 Division (cont.)

Question: Which students took all the subjects?

KETQUATHI		
Mahv	Mamh	Diem
HV01	CSDL	
HV02	CSDL	
HV01	CTRR	
HV03	CTRR	
HV01	THDC	
HV02	THDC	
HV03	THDC	
HV03	CSDL	

MONHOC	
Mamh	Tenmh
CSDL	
CTRR	
THDC	

Mahv
HV01
HV03

KETQUA÷MON

MON (S) ← MONHOC[Mamh]

KETQUA (R) ← KETQUATHI[Mahv, Mamh]

3.7 Division (cont.)

- Division can be expressed by basic relational algebra operations:

$$T \leftarrow R \div S$$

Can be expressed as:

$$T1 \leftarrow \pi_{R^+ - S^+}(R)$$
$$T2 \leftarrow \pi_{R^+ - S^+}(T1 \times S - R)$$
$$T \leftarrow T1 - T2$$

3.7 Division (cont.)

$$T_1 \leftarrow \pi_{R^+ - S^+}(R)$$

T1
Mahv
HV01
HV02
HV03

3.7 Division (cont.)

$$T_2 \leftarrow \pi_{R^+ - S^+}((S \times T_1) - R)$$

T1
Mahv
HV01
HV02
HV03

X

MON
Mamh
CSDL
CTRR
THDC

T1 x MON	
Mahv	**Mamh**
HV01	CSDL
HV01	CTRR
HV01	THDC
HV02	CSDL
HV02	CTRR
HV02	THDC
HV03	CSDL
HV03	CTRR
HV03	THDC

−

KETQUA (R)	
Mahv	**Mamh**
HV01	CSDL
HV02	CSDL
HV01	CTRR
HV03	CTRR
HV01	THDC
HV02	THDC
HV03	THDC
HV03	CSDL

T2	
Mahv	**Mamh**
HV02	

3.7 Division (cont.)

$$T \leftarrow T_1 - T_2$$

T1
Mahv
HV01
HV02
HV03

$-$

T2
Mahv
HV02

$\Rightarrow$

T
Mahv
HV01
HV03

3.7 Division: **Intuition**

$$T \leftarrow R \div S$$

Có thể biểu diễn bằng:

$$T1 \leftarrow \pi_{R^+ - S^+}(R)$$
$$T2 \leftarrow \pi_{R^+ - S^+}(T1 \times S - R)$$
$$T \leftarrow T1 - T2$$

Intuition:

- T1: các MaHV trong KQT
- T2′ ← T1 × S: bắt cặp mọi học viên và mọi môn (= mọi cặp kết quả thi giả sử)
- T2: trừ mọi cặp thi giả sử cho số đã thực sự thi trong R (= còn lại học viên thi chưa đủ môn)
- T: MaHV trong KQT trừ học viên thi chưa đủ môn (= học viên đã thi đủ mọi môn học)

3.8 Aggregate functions and grouping (1)

Old notation:

$$_{G_1,G_2,\ldots,G_n}\Im_{F_1(A_1),F_2(A_2),\ldots,F_n(A_n)}(E)$$

Modern notation:

$$\gamma_{G_1,G_2,\ldots,G_n,F_1(A_1),F_2(A_2),\ldots,F_n(A_n)}(E)$$

Where:

- E is a relational algebra expression
- F_i is the aggregation function, including: $avg(A_i)$, $min(A_i)$, $max(A_i)$, $sum(A_i)$, $count(A_i)$, with A_i is the attribute name
- G_i is the grouping attribute (empty if no grouping)

3.8 Aggregate functions and grouping (2)

- What are the highest, lowest and average scores of the Database (CSDL) subject?

$$\Im_{\max(Diem),\min(Diem),avg(Diem)}(\sigma_{\text{Mamh='CSDL'}}(KETQUATHI))$$

Or:

$$\gamma_{max(Diem),min(Diem),avg(Diem)}(\sigma_{Mamh='CSDL'}(KETQUATHI))$$

KETQUATHI		
Mahv	**Mamh**	**Diem**
HV01	CSDL	7.0
HV02	CSDL	8.5
HV01	CTRR	8.5
HV03	CTRR	9.0
HV05	CTRR	8.0

3.8 Aggregate functions and grouping (3)

- What are highest, lowest, average scores of each subject?

$$_{Mamh}\Im_{\max(Diem),\min(Diem),avg(Diem)}(KETQUATHI)$$

Or:

$$\gamma_{Mamh,max(Diem),min(Diem),avg(Diem)}(KETQUATHI)$$

KETQUATHI		
Mahv	**Mamh**	**Diem**
HV01	CSDL	7.0
HV02	CSDL	8.5
HV01	CTRR	8.5
HV03	CTRR	9.0
HV05	CTRR	8.0

3.9 Renaming relation and attributes

Write concise and clear relational algebra expressions:

- **Split long expressions into smaller units.** E.g.:
  - `T1 ← <expression 1>`
  - `T2 ← <expression 2>`
  - `R ← <expression using T1, t2>`

- **Renaming relations and attributes:**
  - To write subsequences expressions in shorter form.
  - To avoid ambiguous duplicate names of relations and attributes.

3.9 Renaming relation and attributes (cont.)

- Different methods to renaming:
  - Using assignment operator ($\leftarrow$)
  - Using renaming operator (ϱ)
  - Renaming in projection
  - Renaming with aggregation functions and grouping

3.9 Using assignment operator (←)

- **Assignment operator (←):**
  NewR ← R

Example:

KH ← KHACHHANG

3.9 Using renaming operator (ϱ)

- **Renaming operator (ϱ):**

$$\varrho_{NewR}(R)$$

or:

$$\varrho_{NewR(NewAttr1,\ NewAttr2,\ldots)}(R)$$

Example:

$$\varrho_{KH}(KHACHHANG)$$

$$\varrho_{KH(KH_MA,\ KH_HOTEN)}(KHACHHANG)$$

3.9 Renaming in projection

- **Renaming in projection:**

$$\pi_{NewName1:=OldName1,NewName2:=OldName2,\ldots}(R)$$

or:

$$\pi_{NewName1/OldName1,NewName2/OldName2}(R)$$

Example:

$$\pi_{KH_MA:=MAKH,KH_HOTEN:=HOTEN}(KHACHHANG)$$

$$\pi_{KH_MA/MAKH,KH_HOTEN/HOTEN}(KHACHHANG)$$

3.9 Renaming in aggregation functions and grouping

- **Renaming in aggregation functions and grouping:**

$$_{GroupAttr}\Im_{max(Attr)\rightarrow NewName1,min(Attr)\rightarrow NewName2,\ldots}(R)$$

or:

$$\gamma_{GroupAttr,max(Attr)\rightarrow NewName1,min(Attr)\rightarrow NewName2,\ldots}(R)$$

Example:

$$_{Mamh}\Im_{max(Diem)\rightarrow max_diem,min(Diem)\rightarrow min_diem}(KETQUATHI)$$

$$\gamma_{Mamh,max(Diem)\rightarrow max_diem,min(Diem)\rightarrow min_diem}(KETQUATHI)$$

3.9 Renaming relation and attributes (cont.)

- Notes about Join:
  - If attributes are duplicate, you need to specify using `RelationName.AttrName`
    Example: HOCVIEN $\bowtie_{\text{HOCVIEN.MALOP=LOP.MALOP}}$ LOP
  - When renaming, you can rename beforehand or directly in the join expression.
    Example: $\varrho_{\text{HV}}(\text{HOCVIEN}) \bowtie_{\text{HV.MALOP=LOP.MALOP}}$ LOP

BÀI TẬP

Lược đồ CSDL quản lý bán hàng gồm có các quan hệ sau:

KHACHHANG (MAKH, HOTEN, DCHI, SODT, NGSINH, DOANHSO, NGDK)

NHANVIEN (MANV,HOTEN, NGVL, SODT)

SANPHAM (MASP,TENSP, DVT, NUOCSX, GIA)

HOADON (SOHD, NGHD, MAKH, MANV, TRIGIA)

CTHD (SOHD,MASP,SL)

Mô tả các câu truy vấn sau bằng ĐSQH

1. In ra danh sách các sản phẩm (MASP,TENSP) do "Việt Nam" sản xuất có giá từ 30.000 đến 40.000.
2. In ra danh sách các khách hàng (MAKH, HOTEN) đã mua hàng trong ngày 1/1/2007.
3. In ra danh sách các sản phẩm (MASP,TENSP) do "Việt Nam" sản xuất hoặc các sản phẩm được bán ra trong ngày 1/1/2007.
4. Tìm các số hóa đơn mua cùng lúc 2 sản phẩm có mã số "BB01" và "BB02".
5. In ra danh sách các sản phẩm (MASP,TENSP) do "Việt Nam" sản xuất không bán được trong năm 2006.
6. Tìm số hóa đơn đã mua ít nhất hai lần tất cả các sản phẩm do Singapore sản xuất.

Câu 1

- In ra danh sách các sản phẩm (MASP, TENSP) do "Việt Nam" sản xuất có giá từ 30.000 đến 40.000.

$$SANPHAM : ((nuocsx = \text{'Viet Nam'}) \wedge (30.000 \leq gia \leq 40.000))[masp, tensp]$$

hoặc:

$$\pi_{masp,tensp}\sigma_{(nuocsx=\text{'VietNam'})\wedge(30.000\leq gia\leq 40.000)}SANPHAM$$

Câu 2

- In ra danh sách các khách hàng (MAKH, HOTEN) đã mua hàng trong ngày 1/1/2007.

$$\left(KHACHHANG \overset{MAKH}{\bowtie} HOADON : (nghd = \#1/1/2007\#)\right)[makh, hoten]$$

hoặc:

$$\pi_{makh,hoten}\sigma_{(nghd=\#1/1/2007\#)}(HOADON \overset{MAKH}{\bowtie} KHACHHANG)$$

Câu 3

- In ra danh sách các sản phẩm do "Việt Nam" sản xuất hoặc các sản phẩm được bán ra trong ngày 1/1/2007.

$$A \leftarrow SANPHAM : (nuocsx = 'VietNam')[masp, tensp]$$

$$B \leftarrow (SANPHAM \overset{MASP}{\bowtie} CTHD \overset{SOHD}{\bowtie} HOADON : (nghd = \#1/1/2007\#))[masp, tensp]$$

$$C \leftarrow A \cup B$$

Hoặc $A \leftarrow \pi_{masp,tensp} \sigma_{nuocsx='VietNam'}(SANPHAM)$

$$B \leftarrow \pi_{masp,tensp}((\sigma_{nghd=\#1/1/2007\#}(HOADON) \overset{SOHD}{\bowtie} CTHD) \overset{MASP}{\bowtie} SANPHAM)$$

$$C \leftarrow A \cup B$$

Note: Chọn trước khi kết → tối ưu tốt

Câu 4

- Tìm các số hóa đơn đã mua cùng lúc các sản phẩm có mã số "BB01" và "BB02".

$$A \leftarrow CTHD : (masp = 'BB01')[sohd]$$

$$B \leftarrow CTHD : (masp = 'BB02')[sohd]$$

$$C \leftarrow A \cap B$$

Hoặc $A \leftarrow \pi_{sohd} \sigma_{masp='BB01'}(CTHD)$

$$B \leftarrow \pi_{sohd} \sigma_{masp='BB02'}(CTHD)$$

$$C \leftarrow A \cap B$$

Câu 5

- In ra danh sách các sản phẩm do "Việt Nam" sản xuất không bán được trong năm 2006.

$$A \leftarrow \pi_{masp,tensp}\sigma_{nuocsx='VietNam'}(SANPHAM)$$

$$B \leftarrow ((SANPHAM \overset{MASP}{\bowtie} CTHD) \overset{SOHD}{\bowtie} HOADON)$$

$$C \leftarrow \pi_{masp,tensp}\sigma_{(nuocsx='VietNam')\wedge(year(nghd)=2006)}(B)$$

$$D \leftarrow (A - C)$$

Note: Kết trước khi chọn → không tối ưu tốt

Câu 6

- Tìm số hóa đơn đã mua ít nhất hai lần tất cả các sản phẩm do Singapore sản xuất

$$A \leftarrow \pi_{masp}\sigma_{nuocsx='Singapore'}(SANPHAM)$$

$$B \leftarrow \pi_{masp,sohd}\sigma_{sl \geq 2}(CTHD)$$

$$C \leftarrow B \div A$$

Note:
- Nên tập viết rõ phép chia ra
- Nếu đề không yêu cầu rõ thì không cần